

The Clocks:

the full distribution of every duration, by track and parcel value

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Abstract

The model separates a deterministic cost from uncertainty, and the uncertainty channel is identified by dispersion: a level alone cannot tell the two apart. The earlier memos reported means and medians. This memo reports, for every clock the project measures — application to first hearing, first hearing to final action, permit filing to issuance, final action to issuance, issuance to completion, and the developer’s whole clock from first filing to completion — the full distribution, by track, and handles right-censoring with Kaplan–Meier estimates so that recent cohorts count.

The developer’s clock is long and dispersed on every track. Its Kaplan–Meier median is 2,232 days for conditional use, 1,817 for discretionary review, 3,394 for large project authorisations and 1,334 for development permits that never reach the Commission. Among cohorts first filed before 2013, whose spells have had a decade to finish, the coefficient of variation of the completed clock is 0.79, 0.67, 0.52 and 0.95.

About a third of spells never record a completion. Ten years after first filing, 36%, 34%, 48% and 34% of spells on the four tracks are still open. The survival curves therefore never reach their upper quartile, and the upper tail of the developer’s clock — where dispersion lives — is not observed; it is truncated by non-completion, not only by censoring.

Dispersion does not fall consistently with parcel value. Across city-wide quintiles of assessed value per square foot, the CV of the completed developer’s clock goes from 0.67 to 0.81 on the conditional-use track (rank correlation 0.10), from 0.71 to 0.58 on the DR track (-0.40), and from 0.86 to 0.98 off the Commission (0.30). A CV falling with value would be the wedge’s signature. These moments do not show it; they are descriptions, and they do not rule it out.

The hearing is rarely where the time goes. 73% of conditional uses and 68% of DRs are decided at their first hearing; the waits that carry the dispersion are before the hearing and after it, in the permit.

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1 What this memo measures, and why it is a memo of its own

The brief allowed either a new memo or a section of the permits memo. It is a memo of its own because its object is different: the permits memo asks whether projects that reach the Commission differ from similar projects that do not, and answers with means compared within cells; this memo asks for the whole distribution of each duration — the moments a model with an uncertainty channel is estimated on — and its unit of observation, for the Commission tracks, is the case rather than the permit. The code is a stage of the permit-content pipeline (`analyze_permit_content.py` clocks), which it shares its permit table, linkage tiers and development-relevant universe with.

Tracks. A *Commission track* is a case — a case number in the item table — whose first hearing was on a conditional use (including its modification), a discretionary review, or a large project authorisation. The *ministerial or by-right track* is the development-relevant permits (the permits memo’s universe) that no linkage tier connects to the Commission, filed from 1998, when the corpus begins; earlier permits that never recorded a completion would otherwise sit “at risk” for decades and hold up every survival curve. These are 3,205 conditional-use cases, 2,467 DR cases, 150 large project authorisations and 28,498 ministerial permits.

The permit a case’s clock runs through. A case reaches DBI through the permits memo’s tiers T1 (a permit number printed in the minutes) and T2 (the Planning records’ permit field). On the DR track the clock runs through the permit under review — the earliest-filed linked permit, as in the permits memo’s DR clock. Otherwise it runs through the project’s main permit: new construction first, then a site permit, then the largest estimated cost. A case with no linked permit has no permit clock at all (missing, not censored): 45% of conditional-use cases, 94% of DRs and 77% of large project authorisations have one. Many conditional uses are changes of use that need no development permit.

The clocks. Appendix A states each one’s start and end. Two choices matter. *Application* is the opening of the earliest entitlement record for the case (the data-acquisition memo’s entitlement clock) except on the DR track, where it is the DBI filing of the permit under review — the developer’s clock, not the neighbour’s, as the note of 2026-09-11 in that memo now fixes by track. *Final action* is the first hearing at which the item table records a disposition that ends the case (approval, disapproval, taking or not taking DR, withdrawal, adoption, certification, an appeal upheld); a continuance, an intent to approve and “no action” do not.

Parcel value. Assessed value (land plus improvements) per square foot of lot — of building where the roll records no lot area, as for condominium parcels — in the roll year nearest the first filing, binned into city-wide quintiles within that roll year; a case on several parcels takes the median. 87.8% of units are valued. The assessor’s roll begins in 2007, so the 14,593 units first filed earlier are valued at 2007. Proposition 13 holds assessed values at the acquisition price plus two percent a year, so the ranking is by what was paid for a parcel as much as by what it is worth; the price surface in the data-acquisition memo is the alternative, not used here.

Table 1: The units each track is measured on, and how many carry each clock. A spell counts when its start date exists; ‘done’ is the share that has ended by the retrieval date; the rest are censored (still running, or the permit expired or was withdrawn).

Track	application to first hearing		first hearing to final action		permit filed to issued		final action to permit issued		permit issued to completion		first filing to completion		Units
	spells	done	spells	done	spells	done	spells	done	spells	done	spells	done	
conditional use	3,141	100%	3,205	91%	1,436	78%	1,328	78%	1,117	68%	1,440	53%	3,205
discretionary review	2,311	100%	2,467	90%	2,311	83%	2,088	85%	1,912	63%	2,311	53%	2,467
large project authorisation	141	100%	150	81%	115	78%	94	77%	90	74%	115	58%	150
ministerial / by-right	0	—	0	—	28,498	78%	0	—	22,233	67%	28,498	53%	28,498

2 Censoring, and what it cannot fix

A spell still running at the retrieval date (2026-09-11) is right-censored; so is a permit spell that ends in expiry or withdrawal, censored at the exit. Reporting only completed spells understates every duration for recent cohorts, because their long spells have not finished. The Kaplan–Meier estimator counts each censored spell’s exposure up to the point it was last seen, and Figure 1 draws it for every clock and track; Figure 2 gives the hazards of the permit clock and the developer’s clock.

Censoring is not the only thing missing from the upper tail. Table 1 shows that only about half of the developer’s clocks end in a recorded completion, and Table 3 that a third of spells are still open ten years after first filing — far longer than any project is still under construction. Those spells are not running: they are projects abandoned without an exit status, or completions DBI did not record. Kaplan–Meier cannot tell a completion that was never recorded from one that has not happened, so for the developer’s clock its upper quartile is never reached (a dash in the tables) and the completed spells of mature cohorts are the only view of the dispersion. The CV reported for the headline is on those spells: cohorts first filed before 2013, which have had a decade to finish. It describes the projects that finished, and says nothing about the ones that did not.

Kaplan–Meier survival of each clock, by track (censored spells contribute their exposure)

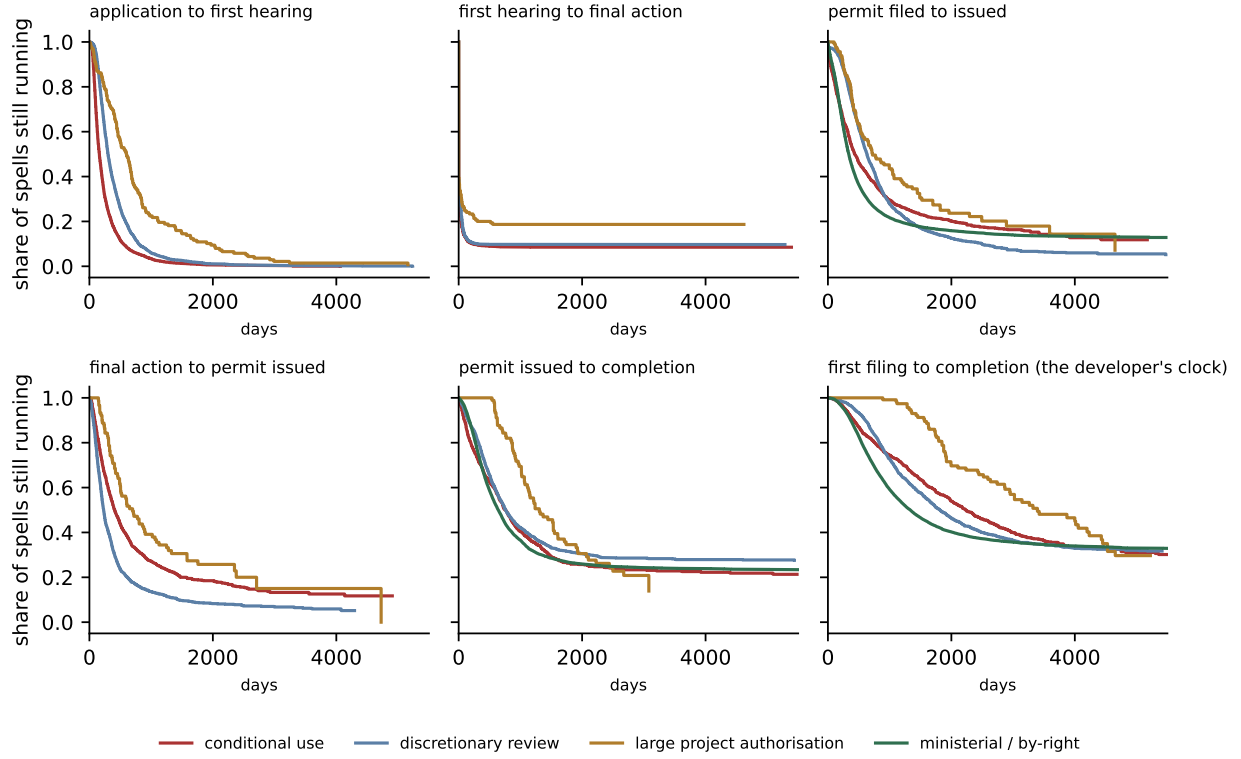


Figure 1: Kaplan–Meier survival of each clock by track, drawn to 5,500 days. The legend gives spells per track. The plateau in the second panel is cases with no final action in the item table; in the last two, spells with no completion recorded.

Hazards: faint 90-day values, bold a one-year centred mean; intervals with fewer than 20 spells at risk not drawn

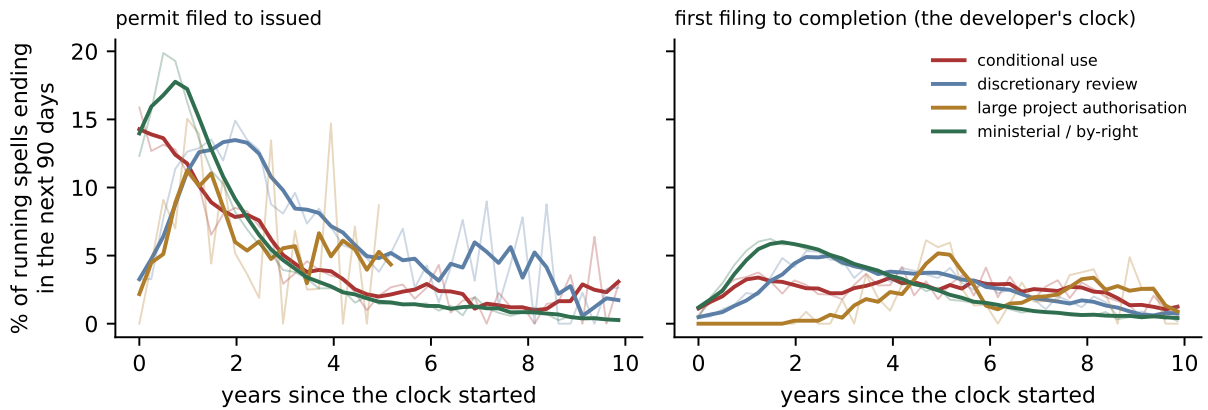


Figure 2: Hazards of the permit clock and the developer's clock by track: the share of running spells that end in the next 90 days.

3 Every clock, by track

Table 2 gives the distribution of every clock on every track: completed-spell percentiles from the 10th to the 99th, the mean, the standard deviation of log days, the coefficient of variation, and the Kaplan–Meier median and 90th percentile beside them. Negative intervals are kept and counted, as in the earlier memos: they enter the quantiles and the mean, and are left out only of the log-scale dispersion.

- **Application to first hearing** is the longest wait at the Commission and the most dispersed: a Kaplan–Meier median of 161 days on the conditional-use track (CV 1.32), 301 on the DR track (CV 1.27) — where the clock starts at the permit filing — and 603 for large project authorisations.
- **First hearing to final action** is short: 73% of conditional uses, 68% of DRs and 75% of large project authorisations are decided at their first hearing. Its CV is high only because the distribution is a spike at zero with a tail of continued cases.
- **Permit filed to issued** has a median of 434, 631 and 722 days on the three Commission tracks against 338 off it — the permits memo’s immortal-time caution applies: a permit reviewed at a hearing had to survive to it.
- **Final action to permit issued** carries negative intervals — a permit issued before the final action (90 conditional-use cases, 70 DR cases) — which are kept.
- **Permit issued to completion** and **the developer’s clock** are the ones non-completion truncates; see the previous section.

Table 2: Every clock, by track: the distribution of completed spells (days) — percentiles, mean, the standard deviation of log days and the coefficient of variation — beside the Kaplan–Meier median and 90th percentile, which count the censored spells’ exposure. ‘Neg.’ is the number of negative intervals, kept in every statistic except the log one. A dash: a cell under 20 observations, or a Kaplan–Meier quantile the curve never reaches.

Clock	Track	<i>n</i>	Neg.	p10	p25	p50	p75	p90	p99	Mean	SD log	CV	KM p50 / p90
application to first hearing	conditional use	3,141	6	63	94	161	301	549	1,681	266	0.85	1.32	161 / 552
	discretionary review	2,311	67	119	182	294	500	790	2,176	399	0.74	1.27	301 / 799
	large project authorisation	141	3	87	279	591	906	1,899	4,393	804	1.11	1.12	603 / 1,969
first hearing to final action	conditional use	2,932	0	0	0	0	7	55	322	20	1.09	3.52	0 / 224
	discretionary review	2,228	0	0	0	0	21	63	222	19	0.94	2.51	0 / 308
	large project authorisation	122	0	0	0	0	5	62	472	29	1.31	3.02	0 / —
permit filed to issued	conditional use	1,116	0	27	118	316	655	1,126	2,900	490	1.39	1.19	434 / —
	discretionary review	1,912	1	204	336	544	853	1,308	2,814	683	0.87	0.82	631 / 2,493
	large project authorisation	90	0	238	366	528	1,049	1,699	3,708	815	0.79	0.93	722 / 4,651
	ministerial / by-right	22,233	0	51	141	263	457	779	2,045	371	1.09	1.08	338 / —
final action to permit issued	conditional use	1,042	90	33	112	270	530	962	2,353	342	1.00	1.88	408 / —
	discretionary review	1,777	70	41	98	190	362	622	1,712	260	0.98	2.06	230 / 1,439
	large project authorisation	72	4	166	282	476	820	1,324	3,293	619	0.77	1.37	700 / 4,726
permit issued to completion	conditional use	763	3	64	155	432	839	1,331	2,593	596	1.30	1.02	762 / —
	discretionary review	1,214	2	102	272	475	763	1,209	2,275	590	1.01	0.87	759 / —
	large project authorisation	67	0	606	853	1,073	1,534	2,083	3,081	1,248	0.45	0.49	1,307 / —
	ministerial / by-right	14,957	7	133	245	415	699	1,078	2,344	547	0.89	0.98	642 / —
first filing to completion	conditional use	763	0	291	517	1,134	1,900	2,668	4,750	1,319	0.89	0.76	2,232 / —
	discretionary review	1,214	1	441	684	1,035	1,617	2,210	3,899	1,231	0.72	0.64	1,817 / —
	large project authorisation	67	0	1,353	1,734	1,995	3,292	4,438	6,629	2,624	0.46	0.50	3,394 / —
	ministerial / by-right	14,970	7	271	440	707	1,122	1,656	3,416	879	0.75	0.80	1,334 / —

4 The headline: the developer’s clock against parcel value

The real-options frame predicts that uncertainty in delay and outcome raises a developer’s filing threshold and re-routes development toward higher-value parcels: uncertainty shifts the threshold differently across parcel values, where a deterministic cost shifts it more nearly uniformly. If the regulatory process imposes more uncertainty on some parcels than others, the dispersion of the clock by parcel value is where it would show first, and a CV falling with parcel value would be the signature of the wedge. Table 3 and Figure 3 give the developer’s clock by track and value quintile.

The signature is not there. On the DR track the CV of the completed clock falls, loosely, from 0.71 in the lowest quintile to 0.58 in the highest (rank correlation across the five quintiles -0.40); on the conditional-use track it has no pattern (0.67 to 0.81, 0.10); off the Commission it drifts up (0.86 to 0.98, 0.30). The permit clock, which completes for most spells, does not line up with it track by track. Its CV goes from 1.12 to 1.23 on the conditional-use track (0.10) and from 0.99 to 0.88 on the DR track (-0.10) — no pattern on either — and from 1.18 to 1.13 off the Commission, where it falls (-0.60) while the developer’s clock on the same track rose. A falling CV appears on one clock of one track at a time, and the two clocks disagree on the ministerial track. The large-project track has too few mature completions per quintile to say anything (40 in all).

This is a description, not an estimate, and it does not reject the prediction. Four things stand between these moments and the model’s object. (i) The completed spells are the projects that finished; if dispersion deters filing or causes abandonment on low-value parcels, the high-dispersion projects there are the ones missing. (ii) Assessed value under Proposition 13 ranks parcels partly by when they last sold. (iii) The quintiles pool every period, and the clock lengthened over time. (iv) The model’s σ is the dispersion a developer faces *ex ante*, conditional on what the developer knows; a cross-sectional CV mixes that with the heterogeneity of the projects themselves — size, type and neighbourhood, which the next section cuts by, one at a time.

Dispersion against parcel value: CV of completed spells, cohorts first filed before 2013; quintiles city-wide within the roll year; cells under 20 not drawn

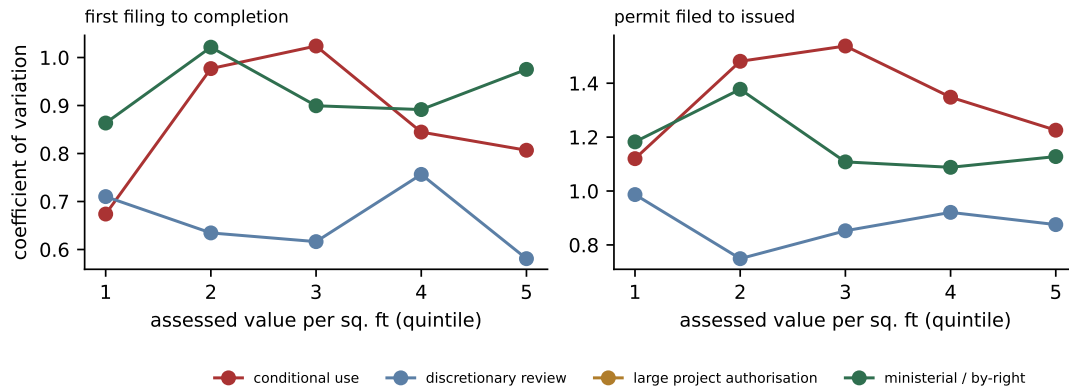


Figure 3: Coefficient of variation of completed spells by parcel-value quintile and track, for cohorts first filed before 2013: the developer’s clock (left) and the permit clock (right).

Table 3: **The developer’s clock — first filing to completion — by track and parcel value.** Value is assessed value per square foot, city-wide quintile of the roll year (1 lowest). Left: Kaplan–Meier quantiles over every spell (days) and the interquartile range over the median; $S(10\text{ yr})$ is the share of spells still running ten years after first filing — no completion recorded. Right: completed spells of cohorts first filed before 2013, which have had a decade to finish — mean, coefficient of variation and standard deviation of log days. Descriptive: these are the moments, not estimates of anything.

Track	Value	Spells	Kaplan–Meier, all cohorts						Completed, first filed before 2013			
			p25	Median	p75	p90	IQR/med.	$S(10\text{ yr})$	n	Mean	CV	SD log
conditional use	1	196	1,021	2,421	—	—	—	0.39	30	1,470	0.67	0.68
	2	220	860	2,691	—	—	—	0.45	57	1,386	0.98	1.22
	3	228	825	2,181	—	—	—	0.39	68	1,310	1.02	1.06
	4	317	1,066	2,338	5,303	—	1.81	0.32	93	1,443	0.84	1.04
	5	288	759	2,231	—	—	—	0.34	79	1,430	0.81	1.06
	all	1,440	960	2,232	—	—	—	0.36	423	1,427	0.79	0.99
discretionary review	1	187	1,051	1,861	—	—	—	0.37	68	1,148	0.71	0.91
	2	242	1,021	2,016	—	—	—	0.36	93	1,271	0.63	0.66
	3	361	940	1,924	—	—	—	0.33	155	1,164	0.62	0.58
	4	566	1,057	2,000	—	—	—	0.38	208	1,158	0.76	0.89
	5	661	943	1,722	—	—	—	0.35	257	1,012	0.58	0.62
	all	2,311	941	1,817	—	—	—	0.34	955	1,113	0.67	0.71
large project authorisation	1	26	1,919	3,422	—	—	—	0.47	5	—	—	—
	2	24	2,519	4,055	—	—	—	0.64	5	—	—	—
	3	20	1,875	3,394	6,174	6,174	1.27	0.46	9	—	—	—
	4	17	—	—	—	—	—	—	—	—	—	—
	5	12	—	—	—	—	—	—	—	—	—	—
	all	115	1,906	3,394	6,174	—	1.26	0.48	40	2,846	0.52	0.48
ministerial / by-right	1	2,680	769	1,482	—	—	—	0.36	592	815	0.86	0.82
	2	3,595	743	1,785	—	—	—	0.41	907	842	1.02	0.79
	3	5,133	678	1,491	—	—	—	0.38	1,540	774	0.90	0.68
	4	6,541	703	1,446	—	—	—	0.36	1,957	813	0.89	0.67
	5	7,231	548	1,128	—	—	—	0.33	2,657	629	0.98	0.71
	all	28,498	637	1,334	—	—	—	0.34	9,373	764	0.95	0.75

5 The other cuts

Tables 4–7 and 8 cut the permit clock and the developer’s clock by proposed units, by the five-year period of first filing, and by the analysis neighbourhood of the project’s permit. Every cell of every clock, with all the statistics of Table 2, is in `clock_moments.csv` (§8); the tables here show the Kaplan–Meier median and 90th percentile and the CV of completed spells, and omit cells under 20 spells. By period the Kaplan–Meier medians of recent cohorts are computed on short follow-up, and their 90th percentiles are mostly not yet reached.

Table 4: Permit filed to issued, by track and proposed units: spells, Kaplan–Meier median and 90th percentile (days), and the median and CV of completed spells. Cells under 20 spells omitted.

Track	Proposed units	Spells	KM p50	KM p90	Done	p50 done	CV done
conditional use	0	305	213	—	248	164	1.20
	1	168	644	—	122	457	1.11
	2–4	307	675	—	241	478	0.96
	5–9	76	616	—	59	376	1.15
	10–19	78	601	3,914	54	337	1.26
	20–49	124	535	3,420	97	355	1.20
	50+	171	482	—	136	376	0.85
	unknown	207	273	—	159	192	1.06
discretionary review	0	109	385	—	73	239	1.07
	1	1,003	574	2,170	862	518	0.82
	2–4	681	709	2,675	551	623	0.75
	5–9	76	729	—	53	498	0.84
	10–19	48	562	2,541	35	385	0.92
	20–49	44	663	2,892	36	598	0.84
	50+	29	336	1,002	25	302	0.56
	unknown	321	730	2,008	277	643	0.84
large project authorisation	20–49	21	1,071	4,651	18	—	—
	50+	60	530	—	51	463	0.65
ministerial / by-right	0	3,013	96	1,203	2,605	76	1.53
	1	11,065	334	3,427	9,068	285	0.96
	2–4	7,934	443	—	5,741	321	0.94
	5–9	1,304	561	—	918	394	0.92
	10–19	827	540	—	601	386	0.90
	20–49	620	391	—	437	248	1.02
	50+	910	198	—	733	152	1.45
	unknown	2,825	286	—	2,130	188	1.43

Table 8: The developer’s clock by analysis neighbourhood (of the project’s permit) and track — CU conditional use, DR discretionary review, LPA large project authorisation, Min. ministerial: spells, Kaplan–Meier median and 90th percentile (days), CV of completed spells. Cells under 20 spells omitted.

Neighbourhood	Track	Spells	KM p50	KM p90	CV done
Bayview Hunters Point	CU	58	2,924	—	0.66

Neighbourhood	Track	Spells	KM p50	KM p90	CV done
	DR	42	—	—	—
	Min.	1,696	1,525	—	0.75
Bernal Heights	CU	34	2,160	—	—
	DR	74	3,340	—	0.84
	Min.	1,181	1,563	—	0.70
Castro/Upper Market	CU	85	3,504	—	0.79
	DR	152	1,917	—	0.62
	Min.	1,229	1,470	—	0.81
Chinatown	CU	34	3,850	—	—
	Min.	278	1,178	—	1.20
Excelsior	DR	49	1,730	—	0.66
	Min.	748	1,403	—	0.68
Financial District/South Beach	CU	66	2,222	—	0.59
	DR	23	2,422	—	—
	LPA	24	2,742	—	—
	Min.	1,778	607	—	1.08
Glen Park	DR	61	1,439	—	0.59
	Min.	542	1,252	—	0.81
Haight Ashbury	CU	30	1,653	—	—
	DR	60	1,759	—	0.49
	Min.	645	1,425	—	0.85
Hayes Valley	CU	54	1,858	—	0.71
	DR	30	—	—	—
	Min.	508	1,938	—	0.86
Inner Richmond	CU	38	2,036	—	0.75
	DR	55	1,190	—	0.67
	Min.	704	1,311	—	0.69
Inner Sunset	CU	31	1,072	3,815	0.90
	DR	91	1,614	—	0.52
	Min.	974	1,308	—	0.74
Japantown	Min.	72	1,329	—	0.52
Lakeshore	Min.	166	525	—	0.68
Lone Mountain/USF	CU	27	1,959	—	—
	DR	28	1,474	—	—
	Min.	440	1,619	—	0.88
Marina	CU	75	1,546	—	0.96
	DR	159	1,820	—	0.59
	Min.	1,032	1,277	—	0.66
Mission	CU	188	2,315	—	0.71
	DR	165	1,703	—	0.79
	Min.	1,797	1,470	—	0.68
Mission Bay	Min.	369	915	—	0.92
Nob Hill	CU	67	2,771	—	0.68
	DR	29	—	—	—
	Min.	436	2,268	—	0.74
Noe Valley	CU	54	3,492	—	0.86
	DR	212	1,810	—	0.63
	Min.	1,724	1,264	—	0.69
North Beach	CU	24	1,463	—	—
	DR	47	3,778	—	—
	Min.	378	1,597	—	0.91
Oceanview/Merced/Ingleside	CU	34	1,493	—	0.57

Neighbourhood	Track	Spells	KM p50	KM p90	CV done
Outer Mission	DR	46	1,512	—	0.58
	Min.	736	1,613	—	0.73
	DR	26	1,462	—	—
	Min.	485	1,469	—	0.72
Outer Richmond	CU	48	1,769	5,358	0.67
	DR	141	1,736	—	0.56
	Min.	1,257	1,527	—	0.81
Pacific Heights	CU	42	1,474	—	0.77
	DR	115	2,549	—	0.67
	Min.	990	1,434	10,247	0.91
Portola	CU	20	2,376	4,466	—
	Min.	415	1,210	—	0.92
Potrero Hill	CU	32	—	—	—
	DR	118	1,953	—	0.60
	Min.	850	1,632	—	0.77
Presidio Heights	CU	28	1,760	—	—
	DR	70	1,378	—	0.51
	Min.	604	1,183	—	0.66
Russian Hill	CU	36	2,180	4,750	0.79
	DR	75	2,302	—	0.65
	Min.	514	2,325	—	0.82
Seacliff	DR	50	2,429	—	0.55
	Min.	244	1,062	—	0.65
South of Market	CU	71	2,439	—	0.62
	DR	50	998	—	0.63
	LPA	39	—	—	—
Sunset/Parkside	Min.	792	1,241	—	0.90
	CU	55	1,284	—	0.72
	DR	101	1,751	—	0.57
Tenderloin	Min.	1,955	1,190	—	0.70
	CU	47	2,464	—	0.56
	Min.	339	1,220	—	0.87
Treasure Island	Min.	50	2,111	—	1.04
Twin Peaks	DR	47	1,900	—	0.56
	Min.	251	1,861	—	0.63
Visitation Valley	DR	23	1,665	—	—
	Min.	316	1,623	—	1.03
West of Twin Peaks	CU	42	2,513	—	0.89
	DR	106	2,598	—	0.56
	Min.	1,525	1,188	—	0.77
Western Addition	Min.	342	1,218	—	0.60

Table 5: First filing to completion, by track and proposed units: spells, Kaplan–Meier median and 90th percentile (days), and the median and CV of completed spells. Cells under 20 spells omitted.

Track	Proposed units	Spells	KM p50	KM p90	Done	p50 done	CV done
conditional use	0	305	1,401	—	170	614	0.88
	1	169	3,615	—	66	1,166	0.83
	2–4	307	2,143	—	162	1,294	0.67
	5–9	77	2,579	—	41	1,117	0.81
	10–19	78	2,178	—	47	1,346	0.70
	20–49	125	2,738	—	70	1,554	0.67
	50+	171	2,632	—	98	1,942	0.47
	unknown	208	1,633	—	109	741	0.72
discretionary review	0	109	3,653	—	43	439	1.17
	1	1,003	1,805	—	538	1,023	0.60
	2–4	681	1,966	—	335	1,175	0.57
	5–9	76	2,004	—	33	894	0.64
	10–19	48	2,054	—	26	948	0.68
	20–49	44	2,493	—	26	1,508	0.63
	50+	29	1,733	5,591	17	—	—
	unknown	321	1,457	—	196	868	0.71
large project authorisation	20–49	21	4,198	—	13	—	—
	50+	60	3,022	—	41	2,106	0.41
ministerial / by-right	0	3,013	708	—	1,848	411	1.08
	1	11,065	1,222	—	6,028	724	0.74
	2–4	7,934	1,644	—	3,667	785	0.75
	5–9	1,304	1,868	—	630	1,125	0.59
	10–19	827	1,857	—	412	1,054	0.59
	20–49	620	1,685	9,281	306	892	0.84
	50+	910	1,274	—	535	831	0.82
	unknown	2,825	1,198	—	1,544	547	0.97

Table 6: Permit filed to issued, by track and first filing: spells, Kaplan–Meier median and 90th percentile (days), and the median and CV of completed spells. Cells under 20 spells omitted.

Track	First filing	Spells	KM p50	KM p90	Done	p50 done	CV done
conditional use	2000–2004	246	390	2,377	215	317	1.26
	2005–2009	245	355	—	194	260	1.40
	2010–2014	232	261	2,107	205	236	1.27
	2015–2019	412	747	—	287	428	0.98
	2020–2024	280	491	—	200	319	0.95
discretionary review	1995–1999	171	380	935	145	351	0.69
	2000–2004	850	565	1,946	692	480	0.74
	2005–2009	525	629	2,493	451	547	0.95
	2010–2014	298	578	2,541	262	548	0.85
	2015–2019	314	902	—	252	785	0.61
	2020–2024	147	882	—	105	726	0.45
large project authorisation	2005–2009	21	908	4,651	18	—	—
	2010–2014	48	537	2,492	44	502	0.76
	2015–2019	23	1,049	—	15	—	—
ministerial / by-right	1995–1999	2,591	191	637	2,339	169	0.96
	2000–2004	7,210	259	2,177	5,875	209	1.22
	2005–2009	4,820	306	—	3,819	244	1.29
	2010–2014	3,826	307	—	3,218	265	1.03
	2015–2019	6,486	498	—	4,853	371	0.86
	2020–2024	2,956	694	—	1,825	416	0.71
	2025–2029	609	226	—	304	146	0.54

Table 7: First filing to completion, by track and first filing: spells, Kaplan–Meier median and 90th percentile (days), and the median and CV of completed spells. Cells under 20 spells omitted.

Track	First filing	Spells	KM p50	KM p90	Done	p50 done	CV done
conditional use	2000–2004	246	1,987	—	154	1,469	0.71
	2005–2009	245	2,013	—	156	1,240	0.81
	2010–2014	232	1,378	—	158	904	0.84
	2015–2019	413	3,504	—	185	1,190	0.60
	2020–2024	282	2,221	—	100	740	0.59
discretionary review	1995–1999	171	1,158	—	94	912	0.69
	2000–2004	850	1,538	—	427	890	0.59
	2005–2009	525	1,474	—	319	1,021	0.69
	2010–2014	298	1,651	—	195	1,250	0.63
	2015–2019	314	3,293	—	133	1,812	0.42
	2020–2024	147	—	—	44	1,336	0.42
large project authorisation	2005–2009	21	3,321	—	15	—	—
	2010–2014	48	2,862	—	37	2,106	0.42
	2015–2019	23	—	—	4	—	—
ministerial / by-right	1995–1999	2,591	807	—	1,592	574	1.07
	2000–2004	7,210	958	—	3,848	562	1.00
	2005–2009	4,820	1,079	—	2,684	628	0.88
	2010–2014	3,826	1,180	—	2,458	760	0.73
	2015–2019	6,486	1,897	—	3,363	1,044	0.55
	2020–2024	2,956	—	—	964	888	0.50
	2025–2029	609	—	—	61	313	0.46

6 The outcome chain: from entitlement to units

The clocks stop at completion; the model’s counterfactual needs the extensive margin as well: conditional on entitlement, what fraction of approved projects — and of approved units — is built, and how long it takes. This section follows every housing project (the principal permit adds at least one unit) from entitlement — the Commission’s first approving action, or for a ministerial permit its filing — through issuance and DBI’s first construction document to the first recorded completion: Housing Production’s completion date on the principal permit, else the first certificate of occupancy in the unit-completion table, else DBI’s completion date. Units delivered are Housing Production’s completed units on the principal permit. Table 9 reads the projects entitled 2005–2014, which have had more than a decade to finish, and adds for every cohort the cumulative incidence of completion within 10 years, with withdrawal, expiry and cancellation as a competing event rather than a censoring: a project that exits does not go on to deliver at the others’ rate.

Table 9: The outcome chain for housing projects (the principal permit adds at least one unit), entitled 2005–2014 (a ministerial permit: filed), by track and city-wide quintile of assessed value per square foot. Shares of projects reaching each stage by retrieval; units delivered are Housing Production’s completed units on the principal permit; years are entitlement to the first recorded completion, among projects that completed. The last column is the cumulative incidence of completion within 10 years over every cohort, with exits as a competing event. Cells under 20 projects are omitted.

Track	Value	<i>n</i>	Issued	Completed	Exited	Open	Units approved	Units delivered	% units	Median years	Done by 10 y
CU	all	141	87	79	17	4	6,583	5,532	84	3.7	68
	Q3	21	90	81	14	5	1,073	1,067	99	3.6	59
	Q4	35	89	80	14	6	965	773	80	3.9	66
	Q5	22	82	77	18	5	539	338	63	4.0	69
DR	all	171	91	78	16	6	386	314	81	2.6	68
	Q1	23	100	70	13	17	30	20	67	2.9	54
	Q2	20	80	55	40	5	31	13	42	2.5	54
	Q3	44	100	93	2	5	171	166	97	2.6	79
	Q4	46	83	78	20	2	78	60	77	2.9	65
	Q5	20	90	70	20	10	49	32	65	2.8	66
LPA	all	33	97	94	3	3	4,037	3,952	98	4.1	64
Ministerial	all	1,791	75	63	28	9	18,531	8,715	47	2.8	56
	Q1	288	74	65	28	7	1,677	391	23	2.9	57
	Q2	279	70	57	33	10	2,128	1,281	60	2.8	48
	Q3	328	72	55	34	11	3,204	1,039	32	2.8	51
	Q4	455	74	62	28	10	2,967	993	33	2.7	54
	Q5	233	73	60	32	9	1,924	460	24	2.5	53

Most entitled housing is built; a large minority is not. Of the housing projects entitled 2005–2014, 79% of conditional uses, 78% of DRs, 94% of large project authorisations and 63% of ministerial permits record a completion; 17%, 16%, 3% and 28% exited, and the rest are still open. Over every cohort, the cumulative incidence of completion within 10 years is 68%, 68%, 64% and 56%: by that measure 32%, 32%, 36% and 44% of entitled projects have not delivered 10 years on. The median time from entitlement to completion, among the mature projects that completed, is 3.7 years for conditional uses, 2.6 for DRs, 4.1 for large projects and 2.8 for ministerial permits.

Units. The mature conditional-use projects were approved for 6,583 net units and Housing Production records 5,532 completed on their principal permits (84%); large project authorisations 4,037 and 3,952 (98%); ministerial permits 18,531 and 8,715 (47%). A completion recorded under another of the project’s permit numbers — an addendum, a second building — is missed, so the

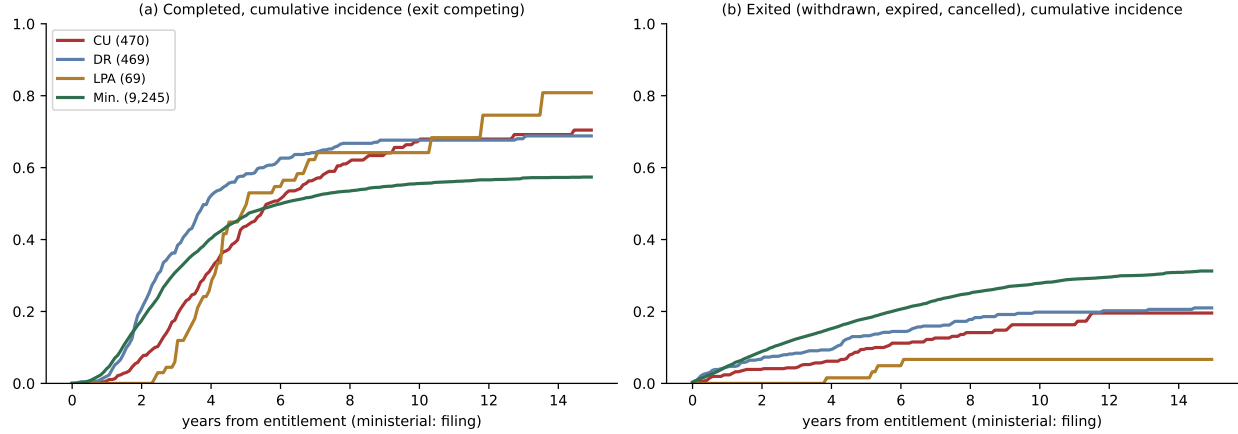


Figure 4: Cumulative incidence of completion (a) and of exit (b) by years from entitlement, housing projects, every cohort; the legend gives the number of projects.

unit shares are lower bounds; the gap is widest on the ministerial track, where the principal permit is the only one this frame follows.

By parcel value. Within tracks the completion share varies across value quintiles without a monotone pattern (81% in the lowest valued quintile shown against 77% in the highest for conditional uses; 65% against 60% for ministerial permits); cells are small on the Commission tracks.

What the chain cannot see. DBI’s first construction document is recorded for 46% of the mature projects, so the chain’s middle link is not reliable and is shown only as a share. Housing Production begins in 2005; completions before it rest on DBI’s own date. And “entitled” is the Commission’s approving action: a project that dies before its hearing, or is never filed, is not in the frame — the deterred-development problem that the risk set is meant to handle.

7 Established, and not established

Established.

1. The full distribution — ten percentiles and moments, and Kaplan–Meier quantiles — of six clocks on four tracks, by units, parcel value, period and neighbourhood, written as a table the model can be fitted to.
2. Dispersion is large on every clock: the CV of the completed developer’s clock is 0.79 (conditional use), 0.67 (DR), 0.52 (large projects) and 0.95 (ministerial).
3. The pre-hearing wait and the permit carry the time; most cases are decided at their first hearing.
4. Across parcel-value quintiles the CV of the developer’s clock does not fall consistently on any track.
5. Of housing projects entitled 2005–2014, 79% (conditional use) to 63% (ministerial) record a completion; the cumulative incidence of completion within 10 years, with exit competing, is 68% to 56%.

Not established.

1. The upper tail of the developer's clock. About a third of spells record no completion within ten years; whether they were abandoned or finished unrecorded is a question for DBI's data, and until it is answered the dispersion reported is that of completed projects.
2. Anything about σ as the model defines it: ex-ante uncertainty conditional on the developer's information. A cross-sectional CV is an upper bound on it only if the cells hold the developer's information fixed, which they do not.
3. The CV-value relation net of period and project size, which needs a joint model, not one-way cuts.
4. Whether the DR track's loosely falling CV is real: five quintiles, a rank correlation of -0.40.
5. The share of approved units built: the unit counts miss completions recorded under a project's other permit numbers, and are lower bounds.

8 Reproducing this

```
python analyze_permit_content.py fetch      # DBI Building Permits, every column
python analyze_permit_content.py report    # permits memo; linkage cache
python analyze_permit_content.py clocks    # this memo
python analyze_permit_content.py outcomes  # the outcome chain
```

`clocks` reads the permit table, the linkage tiers (cached beside the DBI file by `report`), the item table, the Planning records and the assessor's roll, and writes `$MFHR.DATA_ROOT/external/clocks/clocks.parquet` (one row per unit, every clock's duration and event flag and every cut), `clock_moments.csv` (every cell), the three figures, one table file per table and `tables/clocks_macros.tex`. The Kaplan–Meier estimator and the hazard are implemented in the script (`km`, `km_hazard`); the project has no survival library.

A The clocks, defined

- **Application to first hearing:** from the application (the earliest entitlement record of the case, by case number and else by stem; on the DR track, the DBI filing of the permit under review) to the first hearing of the case. Censored at retrieval.
- **First hearing to final action:** from the first hearing to the first hearing recording a disposition that ends the case. Censored at retrieval.
- **Permit filed to issued:** DBI filing to issuance of the case's permit (or of the ministerial permit). Censored at the permit's expiry, withdrawal or other exit, else at retrieval.
- **Final action to permit issued:** from the final action to the permit's issuance. Censored as above.
- **Permit issued to completion:** issuance to DBI's completion date. Censored as above.
- **First filing to completion:** from the earlier of the application and the permit filing to the permit's completion — the developer's clock. Censored as above.